

MATH 301: Homework 2

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Question 1: Section 2.1-10

Section 2.3: q4

Section 2.3: q5

Section 2.3: q6

Section 2.3: q8

Section 2.3: q10

Section 2.4: q2

If $S := \{\frac{1}{n} - \frac{1}{m} | n, m \in \mathbb{N}\}$ find $\inf S$ and $\sup S$.

$$n \in \mathbb{N} \implies 0 < \frac{1}{n} \leq 1$$

$$m \in \mathbb{N} \implies -1 < -\frac{1}{m} \leq 0$$

Finding Supremum

Find the maximum value of $\frac{1}{n}$ and minimum $\frac{1}{m}$.

$$\text{Considering the bounds: } \begin{cases} \implies \frac{1}{n} = 1 \\ \implies -\frac{1}{m} = 0 \end{cases}$$

Note $\sup S = 1$ this statement holds if 1 the smallest upperbound of S .

Consider: $\lim_{m \rightarrow \infty} (1 - \frac{1}{m})$

$$|(1 - \frac{1}{m}) - 1| < \epsilon \iff |-\frac{1}{m} + (1 - 1)| < \epsilon \iff |-\frac{1}{m}| < \epsilon \iff |\frac{1}{m}| < \epsilon \iff \frac{|1|}{m} < \epsilon$$

$$\leftarrow m > \frac{1}{\epsilon} \leftarrow \text{Archimedian Property} \leftarrow \frac{1}{\epsilon} \in \mathbb{R}$$

$$\therefore \lim_{m \rightarrow \infty} (1 - \frac{1}{m}) = 1$$

Consider: $\lim_{n \rightarrow \infty} (\frac{1}{n} - 1)$

$$|(\frac{1}{n} - 1) - (-1)| < \epsilon \iff |\frac{1}{n} + (1 - 1)| < \epsilon \iff |\frac{1}{n}| < \epsilon \iff \frac{|1|}{n} < \epsilon \iff n > \frac{1}{\epsilon}$$

$$\leftarrow \text{Archimedian Property} \leftarrow \frac{1}{\epsilon} \in \mathbb{R}$$

$$\therefore \lim_{n \rightarrow \infty} (\frac{1}{n} - 1) = -1$$

Thus, we can say that the $\sup S = 1$.

Finding Infimum

Find the minimum value of $\frac{1}{n}$ and maximum $\frac{1}{m}$.

$$\text{Considering the bounds: } \begin{cases} \Rightarrow \frac{1}{n} = 0 \\ \Rightarrow -\frac{1}{m} = -1 \end{cases}$$

Consider: $\lim_{n \rightarrow \infty} (\frac{1}{n} - 1)$

Section 2.4: q7

Let A, B be bounded nonempty subsets of $\mathbb{R}$, and let $A + B := \{a + b | a \in A, b \in B\}$.

Prove $\sup(A + B) = \sup A + \sup B$ and $\inf(A + B) = \inf A + \inf B$.

$$\sup A = a_s \implies \forall a \in A, a < a_s$$

$$\sup B = b_s \implies \forall b \in B, b < b_s$$

Observe:

$$\begin{array}{c} (a < a_s) \\ + (b < b_s) \\ \hline a + b < a_s + b_s \end{array}$$

Observe, $a + b$ represent arbitrary elements of $A + B$

$$\text{Therefore, } \sup A + B = a_s + b_s$$

$$\inf A = a_i \implies \forall a \in A, a > a_i$$

$$\inf B = b_i \implies \forall b \in B, b > b_i$$

Observe:

$$\begin{array}{c} (a > a_i) \\ + (b > b_i) \\ \hline a + b > a_i + b_i \end{array}$$

Observe, $a + b$ represent arbitrary elements of $A + B$

$$\text{Therefore, } \inf A + B = a_i + b_i$$

Section 2.4: q9

Let $X = Y := \{x \in \mathbb{R} : 0 < x < 1\}$. Define $h : X \times Y \rightarrow \mathbb{R}$ by $h(x, y) := 2x + y$.

(a) For each $x \in X$, find $f(x) := \sup\{h(x, y) : y \in Y\}$; then find $\inf\{f(x) : x \in X\}$.

(b) For each $y \in Y$, find $g(y) := \inf\{h(x, y) : x \in X\}$; then find $\sup\{g(y) : y \in Y\}$.

Compare with the result found in part (a).

Section 2.4: q14

Section 2.4: q15